RAG Learning

B6.3 Monitoring & Maintaining Health

OCR A GCSE Biology (J247) Gateway — Topic B6: Global challenges

1. Health and Disease

The WHO defines **health** as a state of complete physical, mental and social wellbeing — not just the absence of disease. **Disease** = a condition that affects normal body function. Diseases fall into two broad categories: **communicable** (can be passed from one organism to another) and **non-communicable** (cannot be passed on).

Communicable	Non-communicable
Caused by pathogens (bacteria, viruses, fungi, protists)	Caused by lifestyle, genetics or environment
e.g. flu, measles, TB, malaria, athlete\'s foot	e.g. heart disease, type 2 diabetes, most cancers

2. Pathogens and Communicable Disease

Pathogen type	Examples	How it spreads
Bacteria	Cholera, TB, food poisoning (Salmonella)	Multiply rapidly; release toxins that damage cells/tissues
Viruses	Flu, measles, HIV, COVID-19	Reproduce inside host cells; eventually burst the cell, releasing more viruses

© RAG Learning · Not for redistribution

Fungi	Athlete\'s foot, ringworm	Hyphae grow into skin; release digestive enzymes
Protists (single-celled eukaryotes)	Malaria (carried by mosquitoes)	Often need a vector (e.g. mosquito) to spread between hosts

Methods of disease transmission

- **Direct contact** — touching (e.g. athlete\'s foot, STIs)
- **Air** — droplets from sneezes/coughs (e.g. flu, COVID-19, TB)
- **Water** — contaminated water (e.g. cholera)
- **Food** — contaminated food (e.g. Salmonella, E. coli)
- **Vector** — animal carrier (e.g. mosquitoes carry malaria)
- **Body fluids** — blood, sex (e.g. HIV)

Reducing transmission

- Hand washing, hygiene
- Cooking food properly; clean water supply
- Sterilising medical instruments
- Isolating infected people
- Vector control (e.g. insecticides, bed nets for malaria)
- Vaccination (see below)

3. Body Defences Against Pathogens

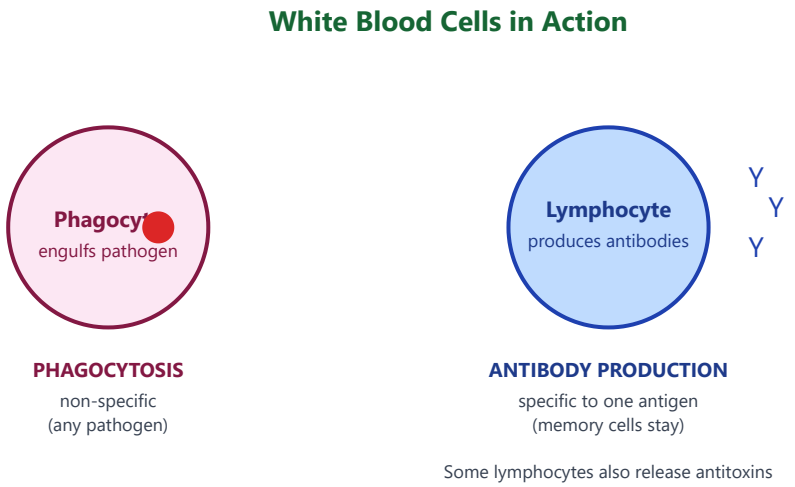
Non-specific (first-line) defences

Barrier	How it works
Skin	Physical barrier; produces antimicrobial chemicals; scab forms over wounds

© RAG Learning · Not for redistribution

Mucus and cilia in airways	Mucus traps pathogens; cilia waft mucus up to throat to be swallowed
Stomach acid (HCl)	Kills pathogens swallowed in food/water
Tears	Contain lysozyme — enzyme that kills bacteria
Inflammation / fever	Increased blood flow brings white blood cells; raised temperature inhibits pathogen growth

The immune system — white blood cells

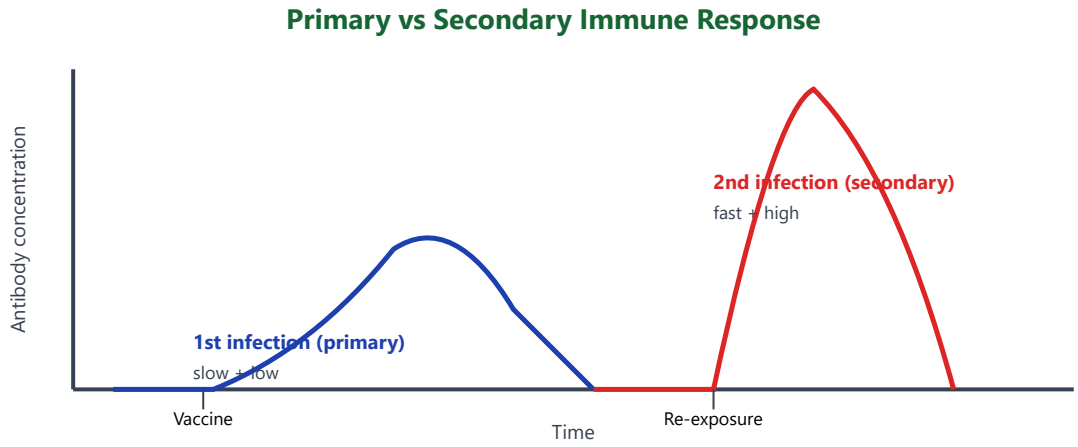


Type of WBC	Action
Phagocyte	Engulfs and digests pathogens (phagocytosis) — non-specific
Lymphocyte	Produces antibodies that bind to specific antigens on the pathogen, marking them for destruction; also produces antitoxins to neutralise toxins; some become "memory cells" giving long-term immunity

4. Vaccination

A **vaccine** contains dead or weakened pathogens (or pathogen fragments) that are introduced into the body. Lymphocytes produce antibodies and form **memory cells**, so if the real pathogen later infects the body, the immune response is **faster and stronger** — the person is **immune**.

Primary vs secondary immune response



Herd immunity

If a high enough proportion of the population is vaccinated, even unvaccinated people are protected because the disease cannot spread easily — this is **herd immunity**. For highly contagious diseases like measles, around 95% vaccination is needed.

5. Antibiotics and Antiviral Drugs

Drug type	Targets	Examples
Antibiotic	BACTERIA only — kill them or stop them dividing	Penicillin, amoxicillin, methicillin
Antiviral	Viruses — slow their reproduction	Used for HIV, herpes, flu
Antifungal	Fungal infections (e.g. athlete\'s foot)	Clotrimazole creams
Painkillers	Treat SYMPTOMS — do not kill pathogens	Paracetamol, ibuprofen, aspirin

© RAG Learning · Not for redistribution

Antibiotics do NOT work on viruses because viruses live inside host cells and use host machinery — there is no bacterial structure to target. Overuse causes antibiotic resistance (see B5.2).

6. Drug Development

New drugs are extensively tested before being released:

Stage	What\'s tested	How
1. Pre-clinical	Toxicity, dosage, efficacy	Tested on cells, then on tissues, then on animals
2. Clinical trials Phase I	Safety; dosage	Small group of healthy volunteers
3. Clinical trials Phase II/III	Safety + efficacy	Larger groups including patients with the condition; usually **double-blind** with **placebo** control
4. Approval and monitoring	Long-term safety	Drug released; effects in real population monitored

> **Placebo** = a "fake" treatment (e.g. sugar pill) used to control for the psychological effect of being treated. **Double-blind** = neither the patient nor the doctor knows who is receiving the real drug — prevents bias.

7. Monoclonal Antibodies

> **Monoclonal antibodies (mAbs)** are identical antibodies produced from a clone of a single B-lymphocyte. They are highly **specific** — they bind to one type of antigen only.

Production

1. A mouse is injected with a target antigen.

2. The mouse\'s lymphocytes produce antibodies.
3. The lymphocytes are extracted and fused with tumour cells (which divide rapidly) to make **hybridoma** cells.
4. Hybridoma cells divide indefinitely, producing huge quantities of identical antibodies.

Uses

Use	How
Pregnancy tests	mAbs detect the hormone hCG in urine — line on stick goes blue if pregnant
Diagnosing disease	mAbs labelled with a fluorescent dye bind to specific tissues — show location of cancers
Cancer treatment	mAbs can carry drug or radiation directly to cancer cells, sparing healthy cells
Research	Identifying specific molecules in tissues / cells

8. Non-Communicable Diseases

Disease	Risk factors
Cardiovascular disease (CVD)	Smoking, high-fat diet, lack of exercise, high blood pressure, obesity, genetics
Type 2 diabetes	Obesity, lack of exercise, poor diet, genetics
Lung cancer / disease	Smoking, pollution, asbestos
Liver disease	Excess alcohol, obesity, hepatitis (viral)
Many cancers	Smoking, UV, radiation, alcohol, viruses (e.g. HPV → cervical cancer), age, genetics

CVD treatments

© RAG Learning · Not for redistribution

- **Lifestyle changes**: stop smoking, healthier diet, more exercise — first line
- **Statins**: drugs to lower cholesterol
- **Stents**: mesh tubes to hold open narrowed coronary arteries
- **Bypass surgery**: replace blocked artery with a vein from the leg
- **Heart transplant**: in severe cases

9. Plant Diseases

Disease	Cause	Symptoms
TMV (Tobacco Mosaic Virus)	Virus	Mosaic pattern on leaves; less photosynthesis → reduced growth
Rose black spot	Fungus	Black spots on leaves; leaves drop early; reduced photosynthesis
Crown gall	Bacterium	Tumour-like growths on stems

Plant defences

Type	Examples
Physical	Cellulose cell walls, waxy cuticle, layers of dead cells (bark)
Chemical	Antibacterial chemicals (e.g. mint), poisons (e.g. caffeine, nicotine)
Mechanical	Thorns, hairs, drooping leaves; mimicry (e.g. passion flower egg-like spots deter insects)

Mineral deficiencies in plants — see B6.2 table.

© RAG Learning · Not for redistribution